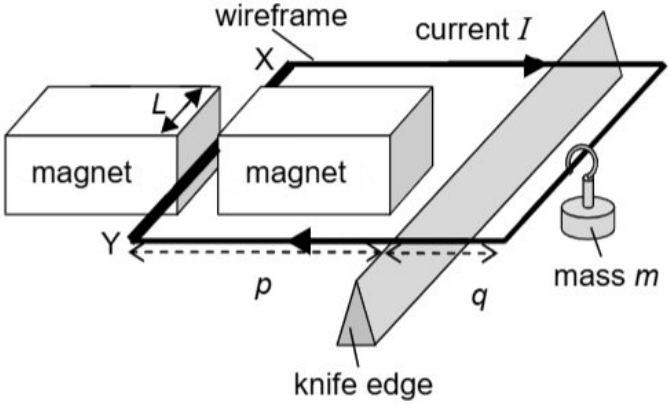
25 An experiment is conducted to determine the magnetic flux density of a pair of magnets using a current balance.

A square wire frame, of negligible mass, is supported horizontally on a knife edge so that XY of the frame lies perpendicular to a pair of magnets of length *L*. Current *I* flows in the wire frame, while a mass *m* is hung at a distance *q* from the knife edge. The wire frame is in equilibrium when XY is at distance *p* from the knife edge.



Which expression gives the magnetic flux density *B* between the magnets?

A $\frac{mgq}{I(p+q)p}$ **B** $\frac{mq}{I(p+q)p}$ **C** $\frac{mgq}{ILp}$ **D** $\frac{mq}{ILp}$